

Research Interests

My current research focuses on 3 areas of Mechanistic Interpretability for Large Language Models (LLMs):

- **Activation Space Geometry (1):** Understanding how concepts and behaviors are represented in the activation space of LLMs. To what extent the Linear Representation Hypothesis and the Platonic Representation Hypothesis hold? [1]
- **Application of Mechanistic Interpretability (2):** Activation Steering, Model Editing, Model Fingerprinting, and other applications. [1, 3, 2]
- **Modular Decomposition (3):** Decomposing LLMs into smaller, interpretable modules that perform specific functions, and studying their interactions. [1, 4, 5]

Publications

*: co-first authorship

- [1] **Hieu M Vu** and Tan Minh Nguyen. “[Angular Steering: Behavior Control via Rotation in Activation Space](#)”. In: *Advances in Neural Information Processing Systems* (2025). **Spotlight (top 3.5%)**.
- [2] Dung V Nguyen*, **Hieu M Vu***, Nhi Y Pham*, Lei Zhang, and Tan M Nguyen. *Activation Steering with a Feedback Controller*. Under review at ICLR 2026. 2025. arXiv: 2510.04309 [cs.LG].
- [3] Ryan Lee*, Dung Viet Nguyen*, **Hieu M Vu***, Lei Zhang, Linh Duy Tran, and Tan Minh Nguyen. *Momentum Steering: Activation Steering Meets Optimization*. Under review at ICLR 2026. 2025.
- [4] Nguyen Hong Son, **Hieu M Vu**, Tuan-Anh D Nguyen, and Minh-Tien Nguyen. “[Jointly Learning Span Extraction and Sequence Labeling for Information Extraction from Business Documents](#)”. In: *2022 International Joint Conference on Neural Networks (IJCNN)*. Oral. IEEE. 2022, pp. 1–8.
- [5] Tuan-Anh D Nguyen, **Hieu M Vu**, Nguyen Hong Son, and Minh-Tien Nguyen. “[A Span Extraction Approach for Information Extraction on Visually-Rich Documents](#)”. In: *Document Analysis and Recognition-ICDAR 2021 Workshops: Lausanne, Switzerland, September 5–10, 2021, Proceedings, Part II 16*. **Best Paper Award**. Springer. 2021, pp. 353–363.

Education

Bachelor Degree, Computer Science (Honours Programme) **2020**
UET - Vietnam National University, Hanoi *GPA: 3.83/4.00 (Rank: 1st/600+)*
• **Highest Ranking Graduate.**

Honours and Awards

ML Alignment and Theory Scholars (MATS) Scholarship **2025**
ML Alignment and Theory Scholars (MATS) Program
A fully-funded, 3-month scholarship for conducting research on ML interpretability and alignment.

Anthropic Fellows Program for AI Safety Research **2025**
Anthropic
Offered fellowship position; unable to participate due to visa constraints.

FPT Smart Cloud Credits for Research, \$2000 **2025**
FPT Smart Cloud

Best Paper Award **2021**
ICDAR 2021, Workshop on Document Images and Language
Paper title: [A Span Extraction Approach for Information Extraction on Visually-Rich Documents](#)

Certificate of Highest Ranking Graduate **2020**
UET - Vietnam National University
Awarded to students graduate with the highest GPA amongst the graduating class.

Top 4 Zalo AI Challenge 2018 - Voice Track (Individual participant) **2018**
Zalo, VNG Corporation
Finished at 4th place on the Private Leaderboard of the Voice Gender/Accent Classification challenge.
[Zalo AI Challenge](#) is an annual Kaggle-like competition hosted by Zalo - one of the biggest tech companies in Vietnam. In 2018, the competition attracted over 700 teams competed in 3 challenges.

Experience

Research Scholar <i>ML Alignment and Theory Scholars (MATS)</i> Fully funded research scholar position under the ML Alignment and Theory Scholars (MATS) program, working on interpretability and alignment of large language models.	Jan 2026 — Present <i>Berkeley, CA, USA</i>
Visiting Scholar <i>National University of Singapore, Advisor: Prof. Tan Nguyen</i> Fully funded visit to NUS to collaborate on research with Prof. Tan Nguyen.	Sept 2025 — Oct 2025 <i>Singapore</i>
AI Research Engineer <i>Torilab</i>	July 2024 – Present <i>Hanoi, Vietnam</i>
• Research on controlling LLM behaviours via activation steering and related intervention methods. • LLM fine-tuning (SFT, DPO), prompt engineering, and large-scale model serving (vLLM, sglang, LitServe). • Developed core framework for customizable chatbots with memory, tool use, and multi-turn interactions. • Additional work: engagement detection, diffusion-based image generation, multi-speaker diarization.	
AI Research Engineer <i>Cinnamon AI</i>	Nov 2018 – May 2024 <i>Hanoi, Vietnam</i>
• Co-creator of kotaemon (24.7k+ ★), an open-source platform for local RAG applications. • Developed RAG-based and LLM-powered applications for industry use cases. • Research in Document Image Understanding: information extraction, cross-lingual pre-training, and model development. • Published peer-reviewed papers; one Best Paper Award at DIL-ICDAR'21. • Led data standardization initiatives and improved internal data management systems. • Mentored engineers and contributed to technical knowledge sharing across teams.	
Undergraduate Research Assistant <i>IOT Lab, UET - Vietnam National University</i>	Aug 2017 — Sep 2019
• Predictive models for wellbore data using machine learning and statistical methods. – Facies/rock type classification; Time-series Analysis; Permeability Regression; Integrated Prediction Error Filter Analysis (INPEFA) curve calculation; Cumulative and Federated Learning.	

Skills

Technical Fields	LLMs, ML Interpretability, RAG, Agentic AI , Information Extraction, Document Understanding, NLP, Computer Vision, Image Processing, Time-series Analysis.
ML/AI Development	Pytorch, Transformers, vLLM , sglang, LangChain, Haystack, LlamaIndex, Tensorflow/Keras, Scikit-learn, OpenCV.
Software Development	Git, Github Action , Docker, CircleCI, DVC.
Mathematics	Linear Algebra, Probability, Statistics, Optimization, Control Theory.
Programming Languages	Python , C/C++, Java, Shell Script.
Industrial Domains	Insurance, Manufacturing, Virtual Companion.
Environments	GCP, AWS, Linux, Windows/WSL.
Natural Languages	Vietnamese (native), English (IELTS 7.5), Japanese (JLPT N4).
Other	Research (Google Scholar), Problem Solving (Leetcode).

Professional Services

Teaching/Mentoring

Cinnamon AI Bootcamp 2020, 2022, 2023

- Mentored groups of 3-4 students.
- Designed syllabus, prepared entrance tests, interviewed candidates.
- Prepared materials and gave lectures on Language Modelling and Transformers.

Reviewer

ICLR 2026, MoFA @ ICML 2025, XLLM-Reason-Plan @ COLM 2025

Conference/Workshop Presentations

- Angular Steering: Behavior Control via Rotation in Activation Space. *NeurIPS 2025* (Spotlight Poster); *Mechanistic Interpretability Workshop at NeurIPS 2025* (Poster).
- A Span Extraction Approach for Information Extraction on Visually-Rich Documents. *ICDAR 2021, Workshop on Document Images and Language* (Oral).

Personal Projects

Kotaemon (24.7k+ ★) - An open-source tool for local RAG application.

Jan 2024 — May 2024

Open-source project, Co-creator



- A local RAG-based tool for chatting with your documents. Built with both end users and developers in mind.
- For end users: A local Question Answering UI for RAG-based QA.
- For developers: A framework for building your own RAG-based QA pipeline.

Data Utility Improvement Experiment for DECAF

Oct 2022 — Nov 2022

Personal research



- A personal research on Causal Inference, Algorithmic Fairness and specifically the paper [DECAF: Generating Fair Synthetic Data Using Causally-Aware Generative networks](#).
- Improved data utility of the DECAF method using alternating graph during synthesis while still achieving similar level of fairness.

Gender/Accent Classification for Vietnamese short voice recordings

Aug 2018 — Sep 2018

Zalo AI Challenge 2018



- Classify the speaker's voice in a recording (typically under 3 seconds) by gender and regional accent.
- **4th place** on the Private Leaderboard, achieved **79.208% accuracy** in 10 days as an **individual participant**.

References

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